

MSC PART 2

Q1: a ; Write a note on kinetics of copolymerization ? 2015

B: Discuss the thermal analysis of polymers in detail .

Q2 a :What is molar mass average? Give its significance in polymer chemistry . 2015

B: Describe principle ,construction and working of Gel Permeation Chromatography (GPC) for molar mass average determination.

Q3: a: Explain Classification of polymers on the basis of structure and application ? 2016

B: Discuss the significance of polymers in daily life.

C: Discuss the kinetics of the copolymerization polymers.

Q4: a: What is molar mass average ? Give different molar mass average . 2016

B: Describe sedimentation method for determination of average molar mass .

Q5: a; What are chain transfer agents in polymers . 2017

B; What is polymerization ? Give different methods of preparation of polymers briefly. 2017+2018+2020

C: Discuss the Kinetics of the free radical polymerization? 2017+2020

Q6: Kinetics of copolymerization? 2017 +2019

Q7: a ; Write down characteristics of Step growth polymerization .Also discuss kinetics of step growth polymerization in the absence of acid catalyst ? 2019

B; Describe sedimentation method for determination of average molar mass .

Q8; Explain classification of polymers on the basis of application and crystallinity ? 2020

Q9: Write a note on Condensation polymerization ? 2020

POLYMER CHEMISTRY

CHEM-443 =CHEM -423 BS REGULAR

SHORT QUESTIONS

- Q1: What is Thermosetting polymers with examples? (2019)
- Q2: Definition of Condensation polymerization ?with Example. (2019)
- Q3: What is Gel Permeation Chromatography ?and its contribution for polymers . (2019)
- Q4: Write classification of polymers on the basis of Thermal behavior? 2020
- Q5: Define Polymerization with example ? 2020
- Q6: Define Terminal model in Kinetics of Copolymerization ? 2020
- Q7: Define Z-average molecular weight? 2020
- Q8: What is the significance of polymers in daily life? 2020
- Q9: Define principle of membrane osmometry? 2020
- Q10 : Define Additional polymerization with example? 2022
- Q11: Polyethylene is a thermoplastic polymer .Justify 2022
- Q12: ? Write distribution order of various average molecular weights 2022

Long Questions

- Q1: Discuss viscosity method for determination of average molar mass of high polymers ? 2019
- Q2; a Explain the Kinetics of Step Growth Polymerization in the absence of acid catalyst ? 2020
- ; Define Chain Growth Polymerization ?What steps involve in Chain growth polymerization.
- Q3: Write Classification of polymerization on the basis of polarity and morphology? 2022



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (6x5=30)

- i. What is Molar mass average? Give its significance.
- ii. What is living polymerization? Explain.
- iii. Differentiate between homo-polymers and copolymers. Give classification of copolymers.
- iv. How does arrangement of molecule affect the properties of polymer?
- v. Describe with examples:
(a) chain transfer agents
(b) polymerization initiators for free radical polymerization
- vi. What are colligative properties? How they help to determine the molar mass of polymer.

Answer the following questions. (3x10=30)

- Q.2** What is condensation polymerization? What are its types? Discuss mechanism and kinetics of condensation polymerization.
- Q.3** Write a detailed note on the determination of number average method.
- Q.4** Discuss thermal analysis techniques used for testing and analysis of polymers.

BS Regular(CHEM-423) (Past Papers)
Polymer Chemistry(CHEM -443)

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M.Sc.-II(Past Papers)

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